

Fig. 4; also a discussion of this point in Appendix D). Consequently, the model was unable to output coherent actions in these trials. This was the case even when we restricted parameter ablation to the middle layers (12-22). Our results suggest that spatial information is widely distributed throughout the model, likely comprised mostly of polysemantic neurons that cannot easily be perturbed without disrupting general model behavior and output coherence.

7 Discussion

Our results demonstrate that representing spatial information in Cartesian coordinates yields the best performance for LLMs in the GWSOT. Cartesian representations consistently outperform Topographic and Textual SIRs across a range of model sizes. It’s interesting, but perhaps expected, that LLMs prefer to process spatial information when it’s presented mathematically, where the x and y dimensions can be operated separately to find the correct orientation, instead of relying on an intuitive understanding of topographical space like humans would. It is also interesting that the JSON variant usually outperforms the Chess Notation SIR, which could be due to the prevalence of such representations in the training data.

We designed a very simple spatial orientation task to be able to test the spatial reasoning of smaller models. While this setting allowed us to isolate the effects of different spatial representations, it also limits the generalizability of our findings to more complex spatial reasoning scenarios. Future work should extend these experiments to larger models and more challenging tasks, such as navigating environments with obstacles, multiple goals, or even three-dimensional spaces. Such studies would help determine whether the advantages of representing space in certain ways persist under more demanding conditions usually found in real-world agentic tasks.

Another promising direction is the exploration of multimodality in spatial reasoning. Currently, our study compares distinct text-based representations of spatial information. Recent works have investigated how combining image and text prompts can impact performance in spatial reasoning tasks [52, 53]. Future work should further explore how integrating visual cues with specific textual descriptions of space might provide complementary advantages in agentic frameworks.

Our analysis of the LLaMA-3.1-8B model provides compelling evidence that even relatively small LLMs develop abstract models of spatial information. We find that all layers contain representations of the grid state up to a linear transformation. Crucially, these representations are mostly dependent on how information is encoded in text, although we observe some degree of generalization, as SIRs from the same class produce more similar spatial encodings. We also find specific units that can represent the position of the agent in the grid by detecting its (x, y) coordinates, as well as detecting whether the agent is near the grid’s boundary. These units are remarkably general, detecting these properties regardless of SIR type, which indicates that they are not encoding prompt-specific

features but the abstract concept of spatial location in the grid. Notably, these units are mostly located in intermediate and deeper layers, matching results from previous research that show higher-level reasoning and conceptual encodings in LLMs can be found in these layers [26–29]. Put together, these findings point to inner models of space that are mostly format-dependent, with some elements encoding abstract spatial features, such as positions in 2D space.

Moreover, we demonstrate the existence of a subset of spatial units—also concentrated in intermediate layers—that can predict whether the selected action will get the agent closer to the goal, showing an understanding of spatial orientation regardless of how information is represented. Remarkably, these units are particularly active even during spatial reasoning tasks outside the grid-world context, suggesting that they can represent spatial information in a very abstract way, regardless of task and prompting. However, it’s important to note that these units may not exclusively process spatial information (i.e. they might not be monosemantic [22, 54]). Furthermore, it is interesting that they are not strictly necessary to perform well on a spatial orientation task, as shown by our ablation experiments. This suggests that they are the most abstract elements of a spatial information processing system that appears to be widely distributed throughout the network. This is supported by the fact that the number of neurons whose activity correlates significantly with action correctness can reach tens of thousands across all model layers for certain SIRs, as shown in Fig. 4. This wider representation of space appears to be less general (as it is SIR-specific) but sufficient to correctly navigate this simple 2D space. It also appears to be comprised of polysemantic neurons, which are necessary to preserve basic output coherence. In the future, it would be interesting to investigate whether the abstract components of the spatial model are necessary when solving harder problems or trying to navigate different kinds of environments. Furthermore, this model of space could be quite different in larger LLMs. We were not able to probe larger models due to constraints in memory and compute, but future work should look for equivalent structures in these larger LLMs to study how these internal representations of space change or stay invariant with scale.

Our work aims to enhance the explainability of LLM-based agentic systems by elucidating how the representation of spatial information influences model behavior. These insights have practical implications for the design of agentic systems. For instance, when deploying LLMs in environments where spatial reasoning is critical, explicitly formatting spatial data in a Cartesian manner can lead to more robust and efficient decision-making. Moreover, understanding the internal structure of spatial representations may inform future strategies for mechanistic analysis and interventions. These might involve reweighing or isolating identifiable units to mitigate unexpected behaviors, as well as using information encoded in these specific groups of neurons to make agentic decision-making more transparent, thus leading to safer and more reliable AI systems.

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